



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Collected SpaceX launch data using REST APIs, web scraping, and structured CSV datasets.
- Performed data wrangling and feature engineering using Python (Pandas, NumPy), including cleaning, encoding categorical variables, and scaling numerical features.
- Conducted Exploratory Data Analysis (EDA) using static visualizations (Matplotlib, Seaborn), SQL queries, and interactive visual analytics.
- Built geospatial visualizations with Folium to analyze launch sites and landing outcomes.
- Developed an interactive Plotly Dash dashboard to explore launch success by site and payload range.
- Trained and evaluated multiple classification models (Logistic Regression, SVM, Decision Tree, KNN) using GridSearchCV for hyperparameter tuning.

Summary of results

- Identified KSC LC-39A as the launch site with the highest number of successful launches and the highest success rate
- Found that payload ranges between ~2,000 kg and 6,000 kg were associated with the highest launch success rates.
- Observed that newer Falcon 9 booster versions (FT, B4, B5) consistently achieved higher success rates than earlier versions.
- Among the evaluated machine learning models, the Decision Tree classifier achieved the best cross-validated accuracy (~89%), outperforming Logistic Regression, SVM, and KNN.
- Overall results demonstrate that launch site, payload mass, and booster version are strong predictors of mission success.

Introduction

Project background and context

- SpaceX aims to significantly reduce the cost of space travel by reusing rocket boosters, making launch success a critical factor in operational efficiency.
- Understanding the factors that influence successful Falcon 9 launches and landings is essential for predicting mission outcomes and improving launch reliability.
- This project analyzes historical SpaceX launch data to uncover patterns related to launch site, payload mass, orbit type, and booster version.

Problems you want to find answers

- What factors most strongly influence the success or failure of Falcon 9 launches?
- How do launch sites, payload mass, and booster versions affect launch outcomes?
- Can machine learning models accurately predict launch success based on historical data?
- Which classification model performs best for predicting SpaceX launch outcomes?

Section 1

Methodology

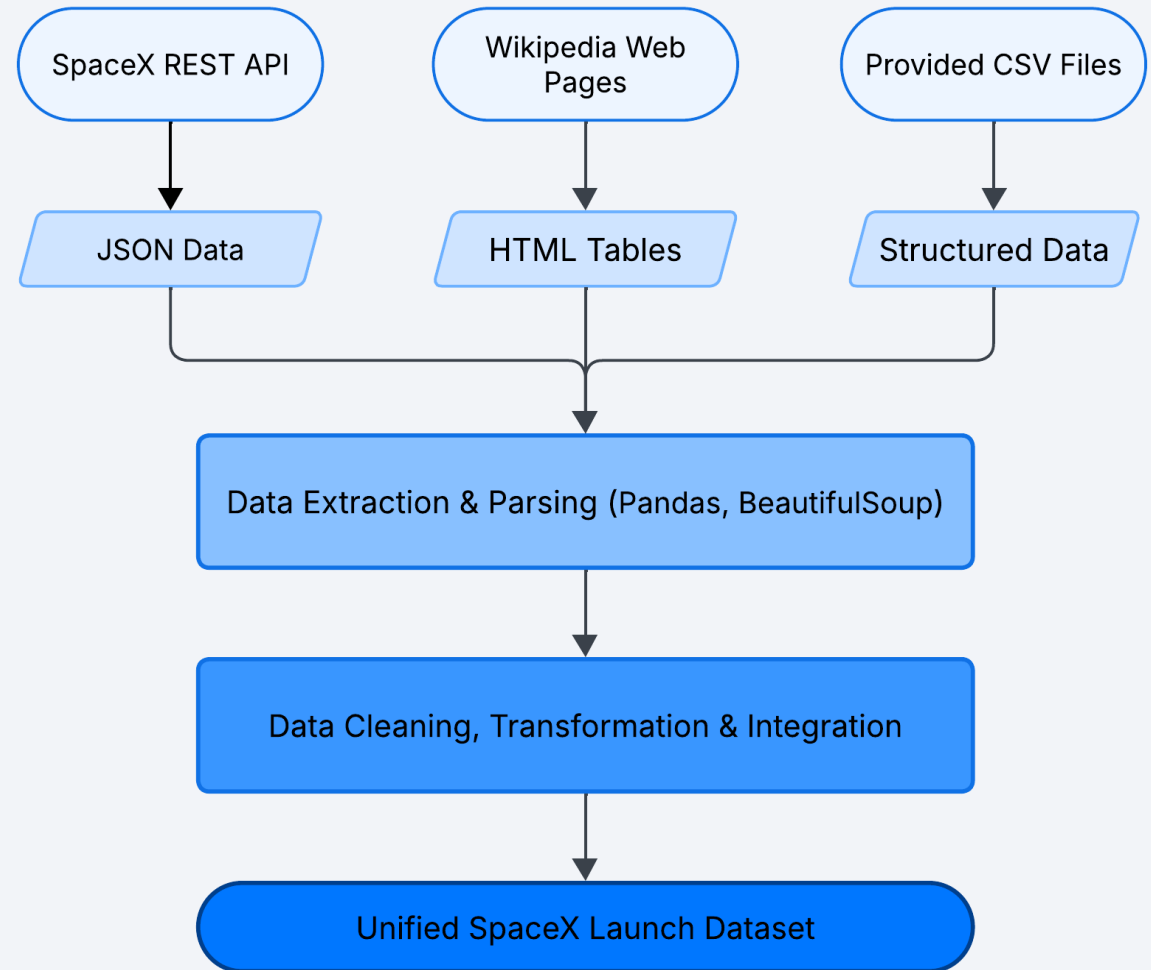
Methodology

Executive Summary

- **Data collection:** data was collected from REST APIs, web scraping from Wikipedia, and provided CSV datasets.
- **Data wrangling:** cleaned and preprocessed the raw data using Python (Pandas, NumPy); handled missing values, removed irrelevant columns, and converted categorical variables into numerical representations; engineered a binary launch success variable (Class); standardized numerical features.
- **Exploratory data analysis (EDA):** performed EDA using Matplotlib and Seaborn; used SQL to aggregate, filter, and analyze launch success rates; combined SQL and Python-based EDA to validate insights across different analytical approaches.
- **Interactive visual analytics:** built interactive geospatial maps using Folium to analyze launch site locations and landing outcomes; developed an interactive Plotly Dash dashboard with dropdowns and sliders to explore launch success by site and payload range in real time.
- **Predictive analysis:** built and evaluated multiple classification models, including Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN); applied GridSearchCV for hyperparameter tuning and cross-validation; evaluated models using accuracy scores and confusion matrices

Data Collection

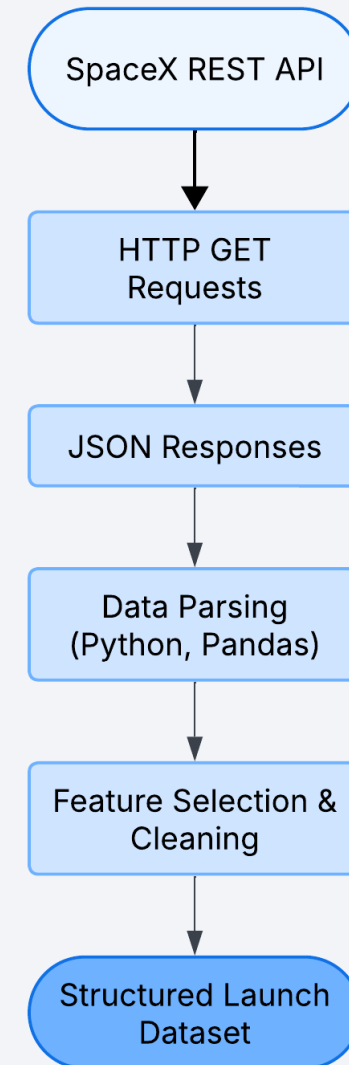
- Collected SpaceX launch data from multiple heterogeneous sources to ensure completeness and reliability.
- Used REST APIs to retrieve structured launch and rocket information programmatically.
- Performed web scraping on publicly available Wikipedia pages to extract historical launch records and payload details.
- Integrated provided CSV datasets containing curated SpaceX launch information.
- Combined all data sources into a single unified dataset for downstream analysis.



Data Collection – SpaceX API

- Collected historical SpaceX launch data using publicly available SpaceX REST API endpoints
- Retrieved launch, rocket, payload, and landing outcome information in JSON format
- Performed multiple API calls and joined responses using unique identifiers (e.g., launch IDs, rocket IDs)
- Parsed and transformed JSON responses into structured tabular data using Pandas

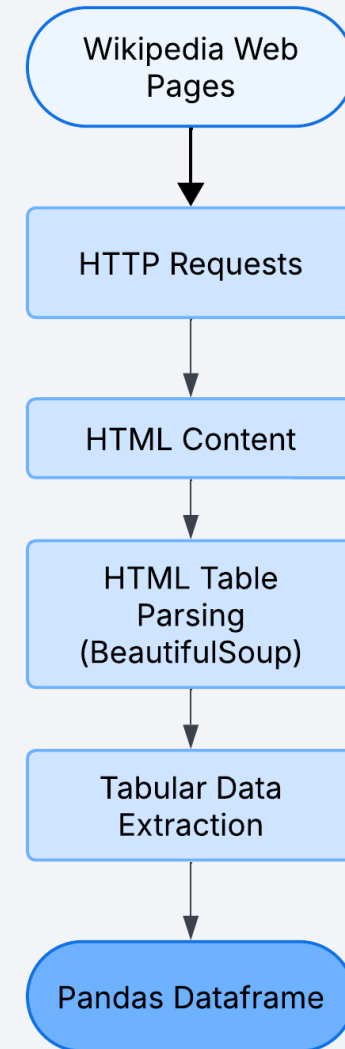
Notebook: [spacex-data-collection-api.ipynb](#)



Data Collection - Scraping

- Collected additional SpaceX launch information by scraping publicly available Wikipedia pages
- Extracted launch records and landing outcome details from structured HTML tables
- Used requests and BeautifulSoup to parse HTML content
- Converted scraped tables into Pandas DataFrames for further processing

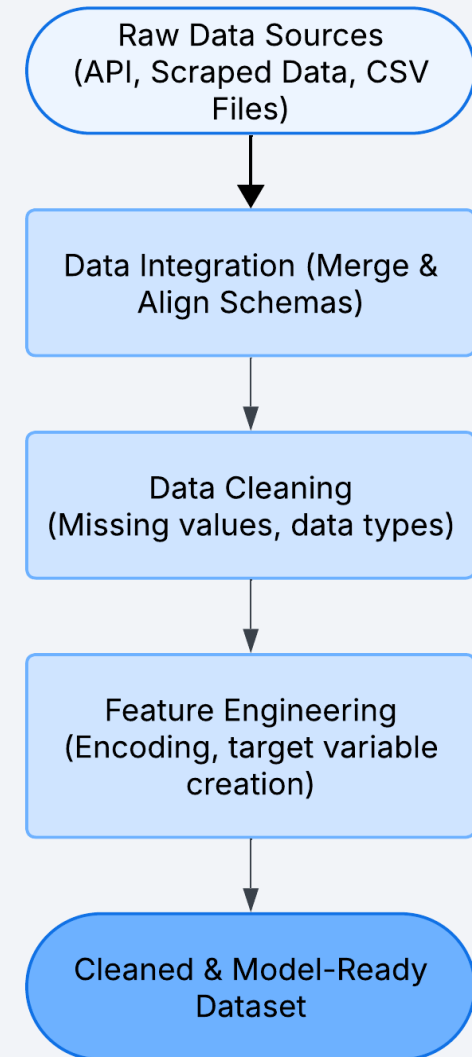
Notebook: [spacex-webscraping.ipynb](#)



Data Wrangling

- Merged data collected from SpaceX API, web scraping, and provided CSV files into a unified dataset
- Handled missing values and inconsistent entries across multiple data sources
- Converted categorical variables into numerical features (one-hot encoding)
- Created a binary target variable (Class) to represent landing success
- Selected relevant features for exploratory analysis and predictive modeling

Notebook: [spacex-data-wrangling.ipynb](#)



EDA with Data Visualization

To identify patterns and relationships between launch success, time, payload, orbit type, launch site, and operational experience, several charts were plotted:

- Line plots were used to analyze temporal trends in launch success rates
- Bar charts were used to compare average success rates across orbit types, enabling direct comparison between missions with different orbital difficulty levels.
- Scatter plots were used to explore relationships between payload mass, flight number, orbit type, and launch site.
 - Coloring points by mission outcome (Class) allowed visual separation of successful and failed launches. These plots helped assess whether increased experience and payload capacity were associated with higher success rates.
- Notebook: [spacex-eda-visualizations.ipynb](#)

EDA with SQL

SQL was used to explore, filter, and aggregate SpaceX launch data in order to identify patterns in launch sites, payload characteristics, landing outcomes, and booster performance.

- Queried unique launch sites and filtered missions using pattern matching (DISTINCT, LIKE)
- Retrieved specific subsets of launches based on launch site, date range, and payload constraints
- Computed aggregate metrics such as total and average payload mass using SUM() and AVG()
- Analyzed landing outcomes frequency using GROUP BY and COUNT
- Identified first successful ground landing using date aggregation (MIN)
- Used subqueries to find booster versions associated with maximum payload mass
- Performed time-based filtering (year/month extraction) to analyze mission outcomes in 2015
- Ranked landing outcomes over time using grouped counts and ordering

Build an Interactive Map with Folium

Folium was used to create an interactive map to analyze the geographic distribution of SpaceX launch sites and launch outcomes.

Map objects created:

- **Markers & circle markers** for launch site locations to provide an intuitive spatial overview of all launch sites
- **Color-coded markers** to represent successful vs failed launches to allow for immediate visual comparison
- **Marker clusters** to manage overlapping launches and reduce visual clutter
- **Polylines** to help quantify proximity to coastlines and populated areas
- **MousePosition plugin** to dynamically inspect latitude and longitude values

Notebook: [spacex-folium-visualizations.ipynb](#)

Build a Dashboard with Plotly Dash

An interactive dashboard was developed using Plotly Dash to explore launch performance across different launch sites, payload ranges, and booster versions.

- **Pie charts** showing success vs failure (all sites and individual sites) to provide an immediate visual comparison of launch success rates across different launch sites and to highlight performance differences between locations.
- **Scatter plots** of payload mass vs launch outcome, colored by booster version, to explore the relationship between payload mass and launch success, and to assess whether specific booster versions are associated with higher success rates.
- **Launch site dropdown selector** to dynamically filter the dashboard by launch site, making it easier to perform focused, site-level analysis without cluttering the visuals.
- **Payload mass range slider** for interactive filtering of launches by payload mass, helping users identify trends or thresholds where success rates may change.

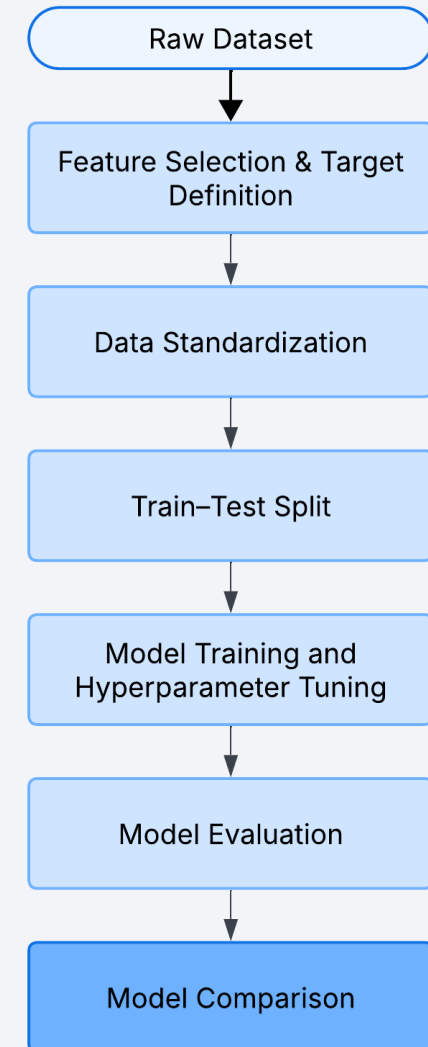
Notebook: [spacex_dash_reference.ipynb](#)

Predictive Analysis (Classification)

Model development & evaluation

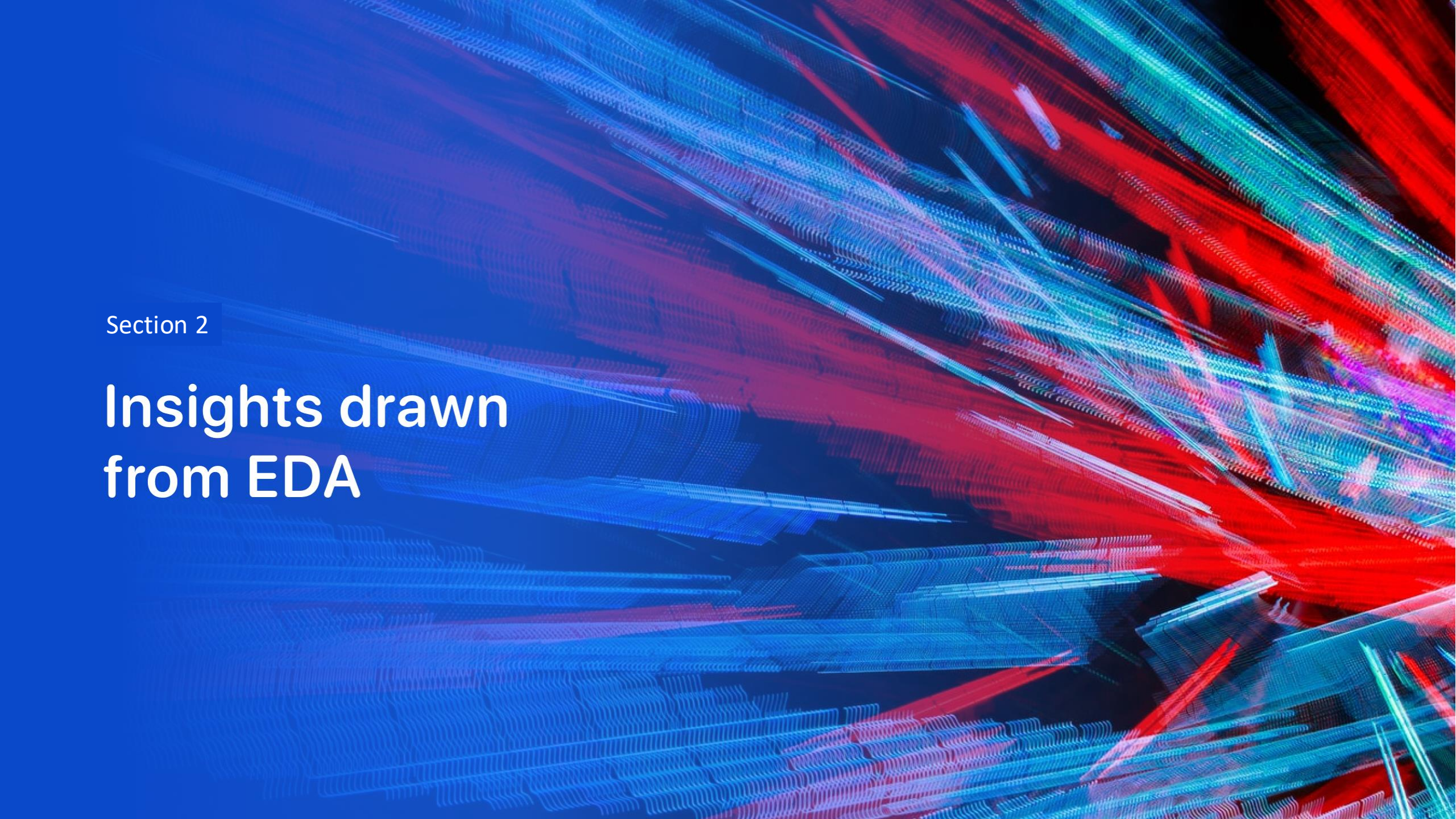
- Data preparation:
The target variable (launch success/failure) was extracted, and all numerical features were standardized using StandardScaler to ensure fair model comparison.
- Train-test split:
The dataset was split into training (80%) and test (20%) sets
- Model training:
Four classification models were implemented: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- Hyperparameter tuning:
Each model was optimized using GridSearchCV with 10-fold cross-validation, allowing systematic exploration of hyperparameters and reducing overfitting.
- Model evaluation:
Performance was assessed using validation accuracy, test accuracy, and confusion matrices to understand both overall accuracy and error distribution.

Notebook: [spacex-ml-predictions.ipynb](#)



Results

- Exploratory Data Analysis Results:
 - Launch success rate increased steadily over time.
 - Launch site and orbit type influence success probability.
 - Payload mass alone is not a decisive predictor.
 - Certain orbit types (LEO, ISS, SSO) show higher success rates.
- Interactive Analytics Demo
 - Dashboard enabled filtering by site and payload.
 - Visual patterns revealed mission-specific success behavior.
 - Geographic mapping illustrated launch site proximity factors.
- Predictive Analysis Results
 - Decision Tree performed best (~0.89 validation accuracy).
 - Other models achieved comparable performance (~0.84–0.85).
 - Test accuracy ~0.83 confirms reasonable generalization.

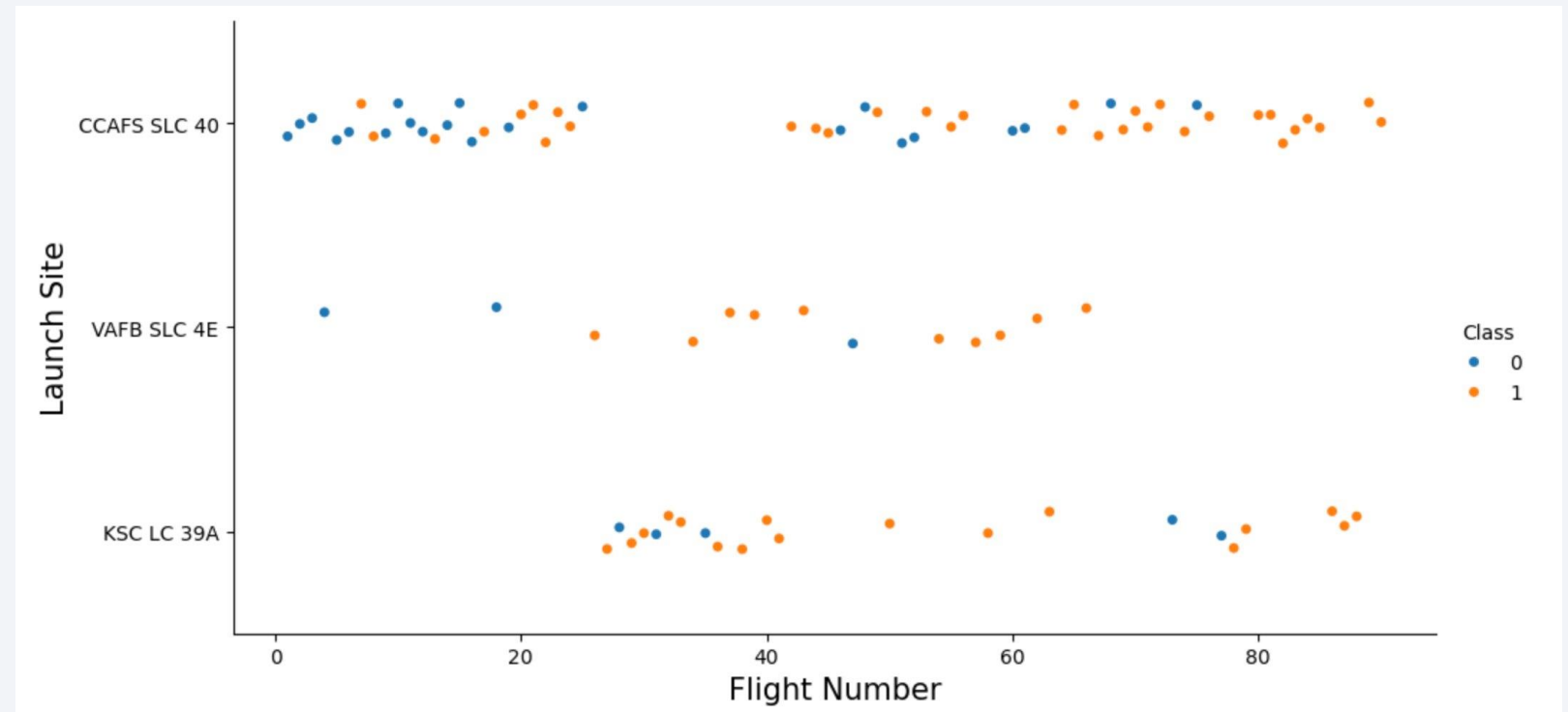
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

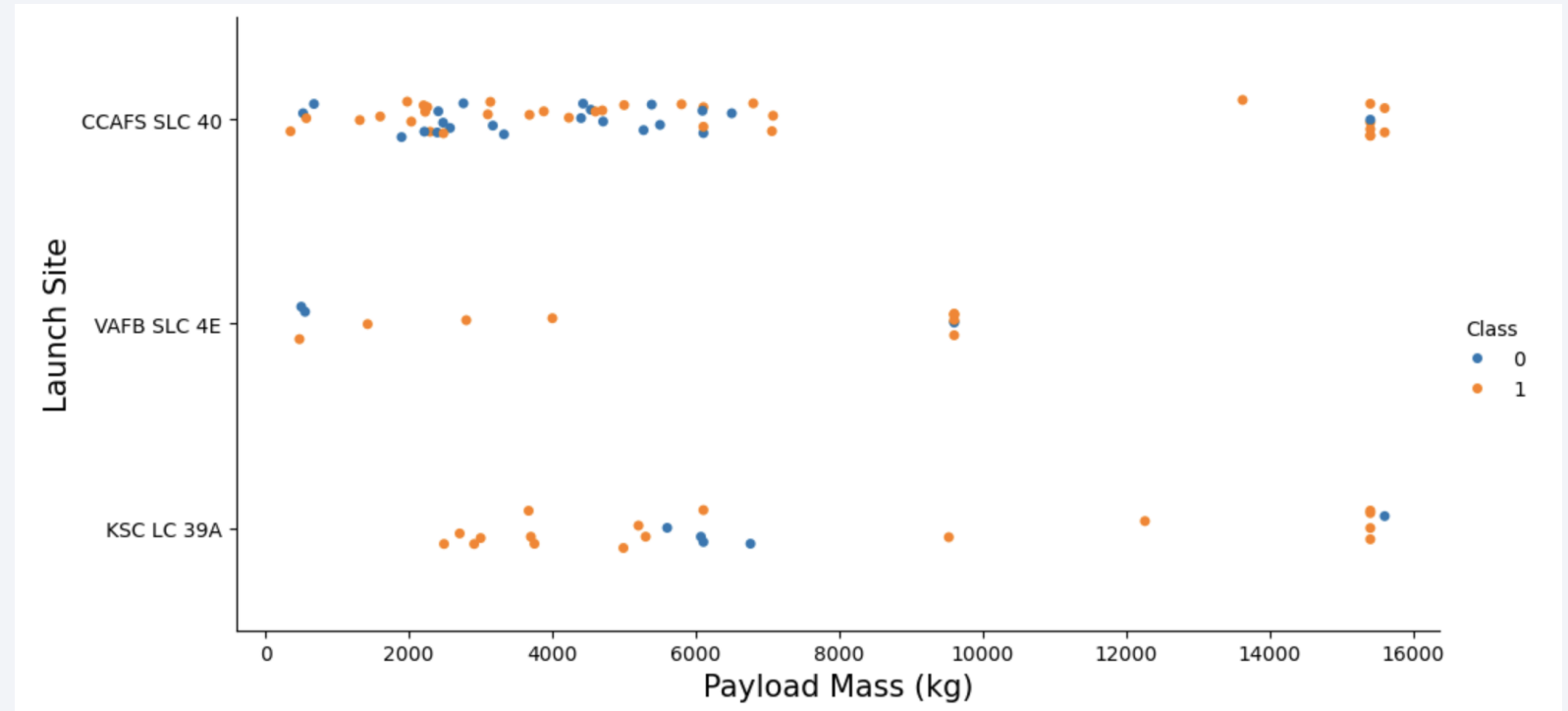
Flight Number vs. Launch Site

- Later flight numbers show a higher concentration of successful landings (Class = 1).
- Success rate improves over time across most launch sites.
- CCAFS SLC-40 shows both early failures and later consistent successes.
- Indicates operational learning and technological improvement over time.



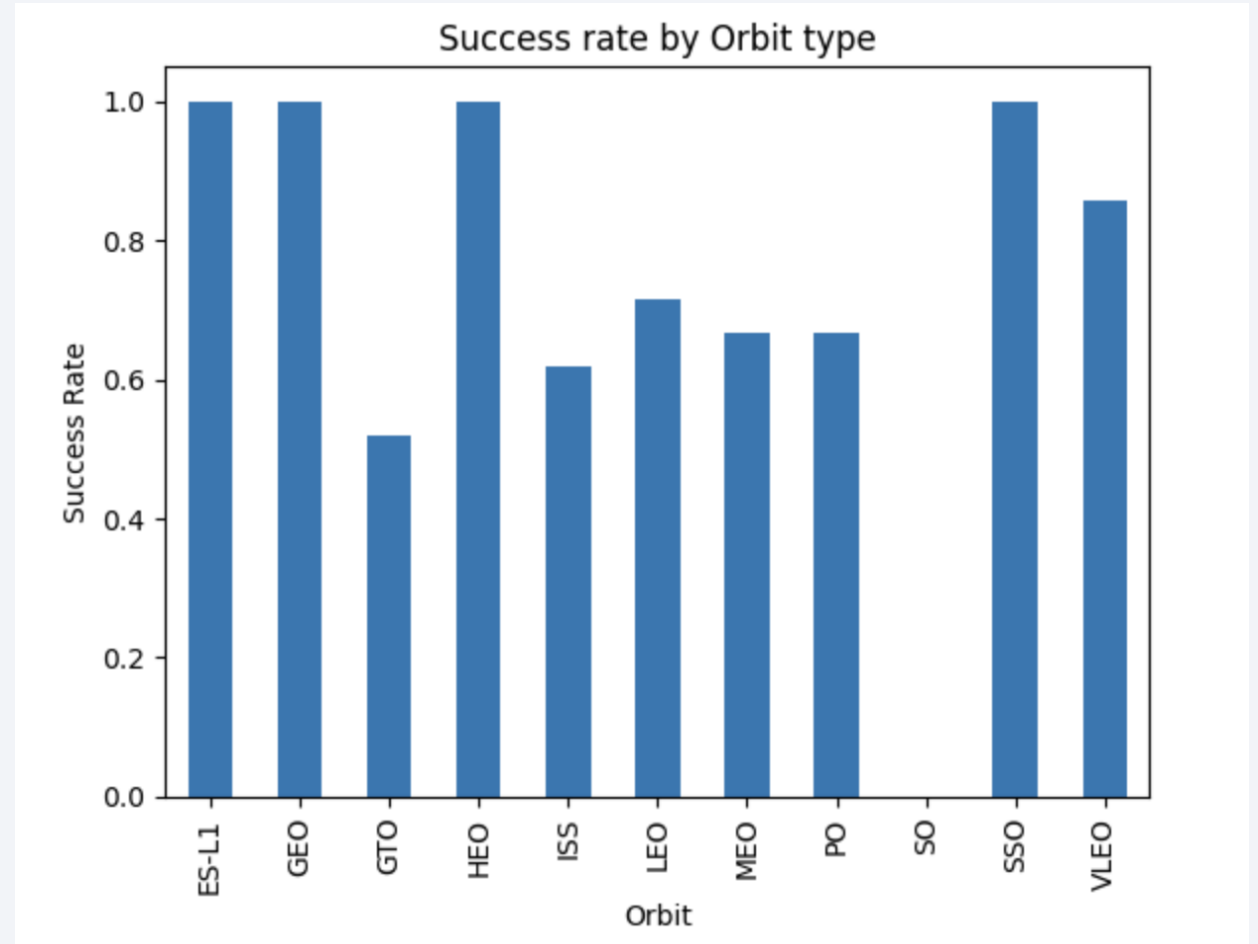
Payload vs. Launch Site

- VAFB SLC-4E does not launch very heavy payloads ($>10,000$ kg).
- CCAFS SLC-40 handles a wide range of payload masses.
- KSC LC-39A includes several high-mass missions with successful outcomes.
- Launch site capabilities vary depending on mission requirements.



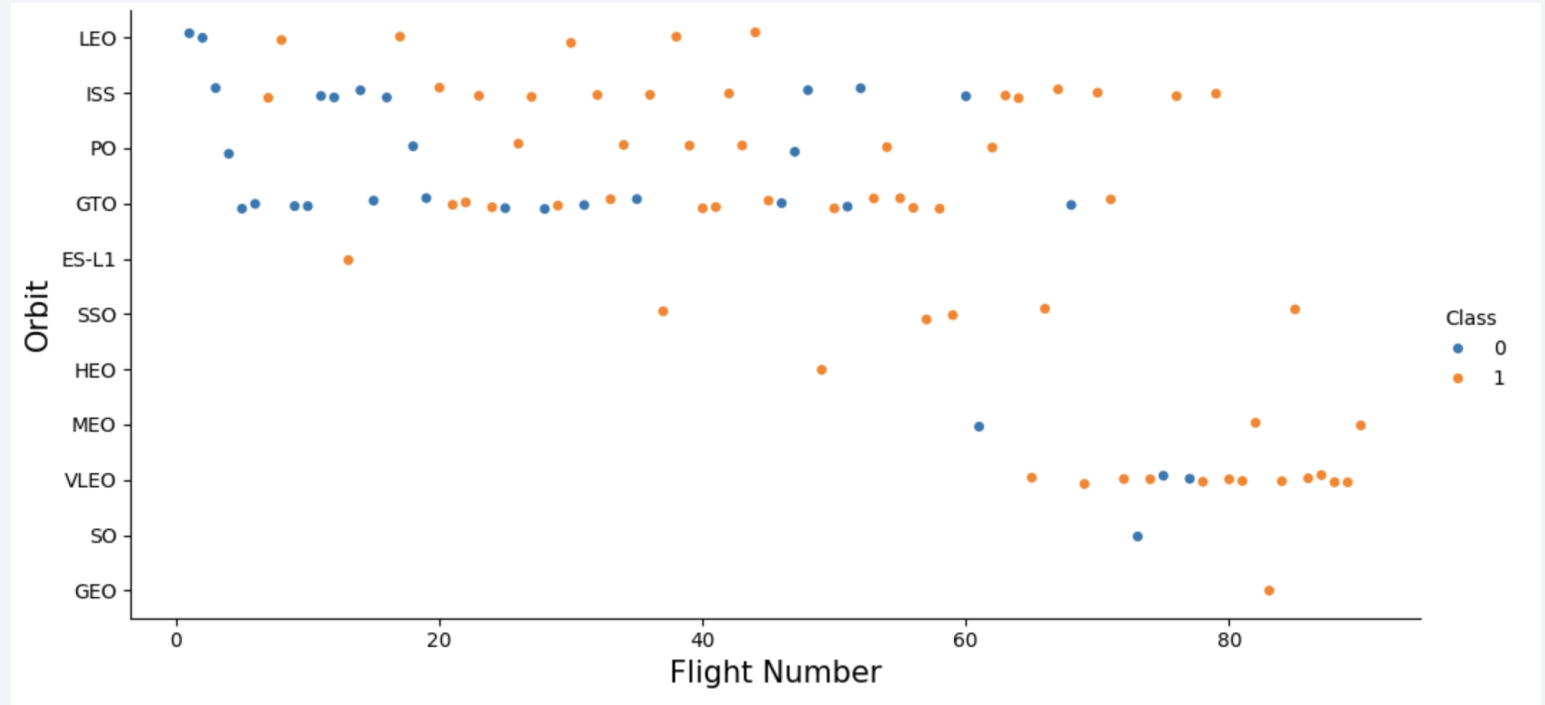
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, and SSO show very high success rates.
- LEO, MEO, and PO have moderate but stable success rates.
- GTO presents mixed results, with both successes and failures.
- Orbit type appears to influence landing success probability.



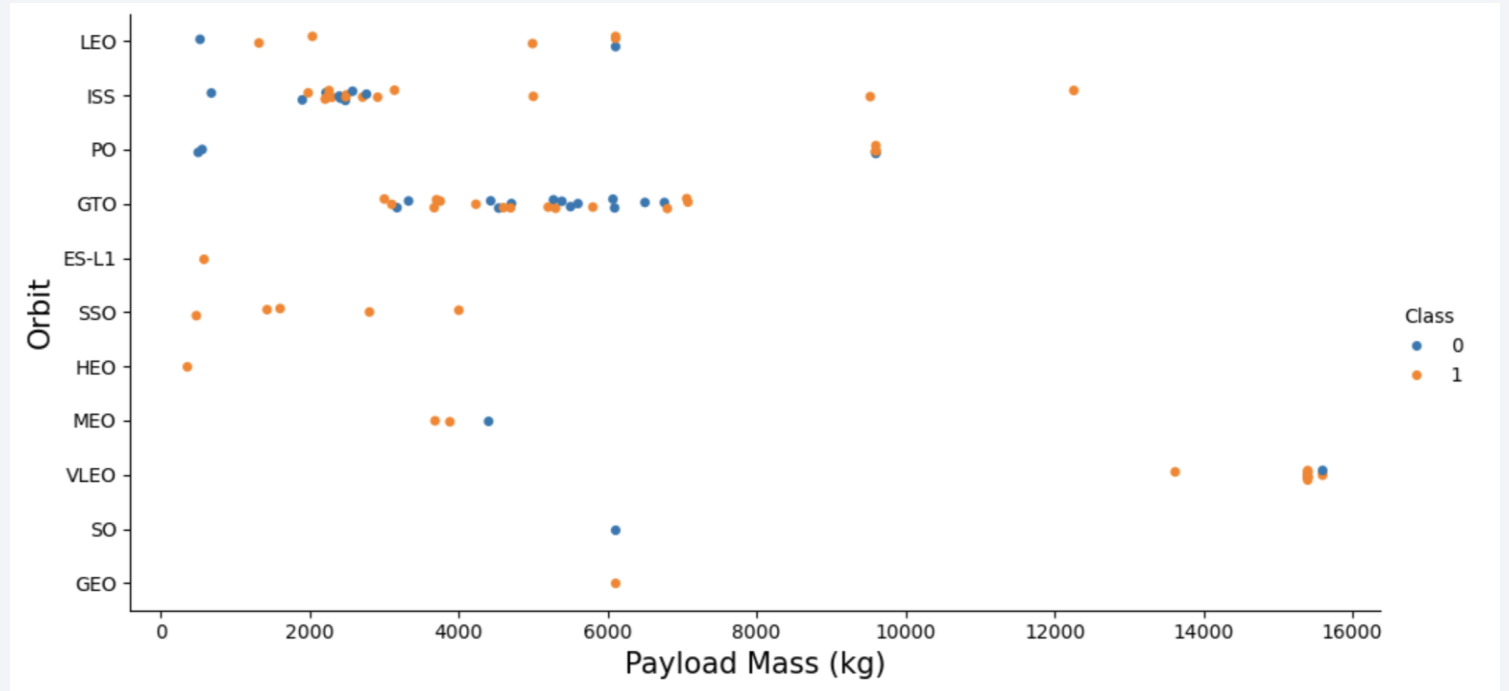
Flight Number vs. Orbit Type

- In LEO missions, success increases with flight number.
- GTO missions show mixed outcomes across flight history.
- Later missions in several orbit types show higher success frequency.
- Suggests experience improves performance in certain orbit categories.



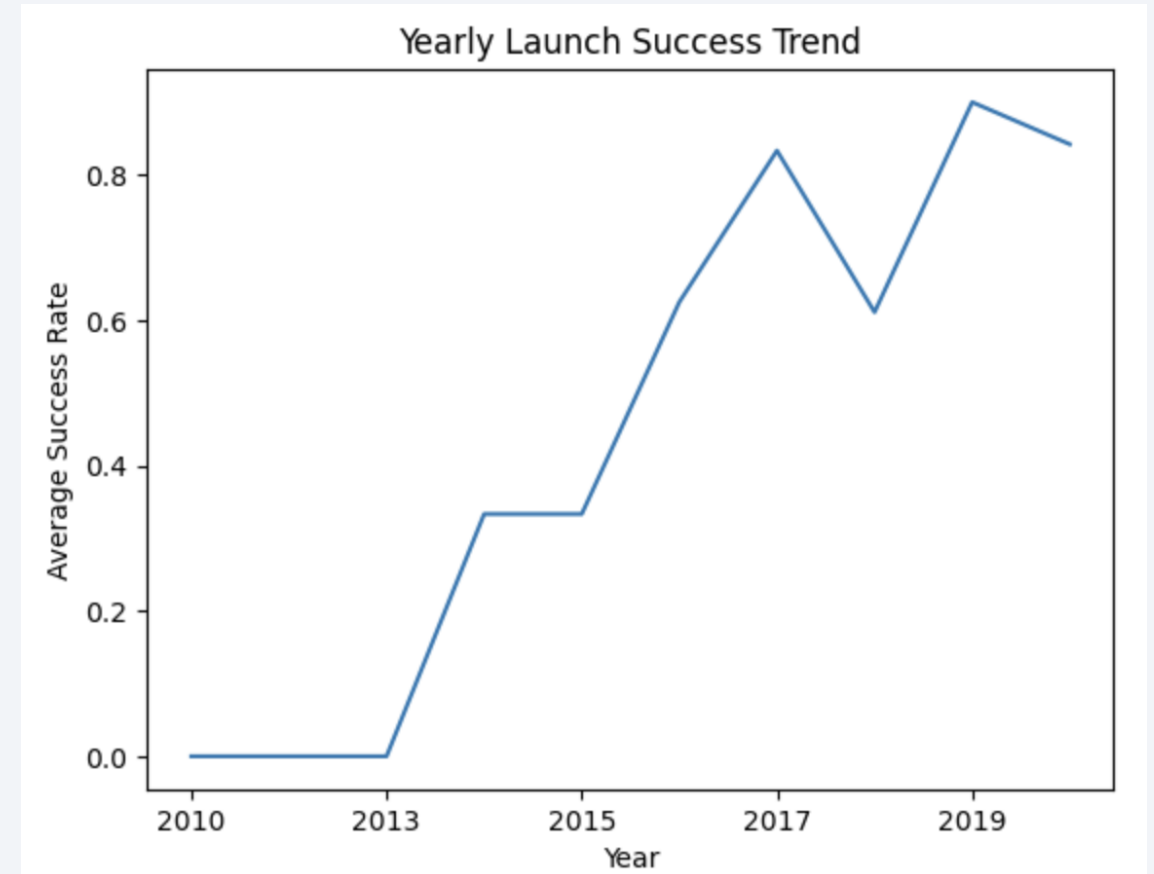
Payload vs. Orbit Type

- Heavy payload missions (> 12,000 kg) are mostly associated with VLEO and certain LEO missions.
- For LEO and ISS, heavier payloads often correspond to successful landings.
- GTO missions show no clear separation between success and failure based on payload mass.
- Payload mass alone is not a decisive predictor of landing outcome.



Launch Success Yearly Trend

- Success rate remained very low before 2013.
- Strong upward trend from 2014 onward.
- Peak performance observed around 2019.
- Indicates continuous technological and operational improvement over time.



All Launch Site Names

- Launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A
- The dataset contains three distinct launch sites.
- These sites represent the primary operational locations for Falcon 9 missions.

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- All returned records belong to CCAFS LC-40.
- Early missions show “Failure (parachute)” or “No attempt” landing outcomes.
- Indicates early-stage landing development phase.

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload mass = 45,596 kg
- NASA CRS missions contributed significantly to total payload mass.
- Demonstrates long-term commercial partnership impact.
- CRS missions form a substantial portion of operational launches.

```
%sql SELECT SUM("PAYLOAD_MASS__KG_") AS total_payload_mass FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db  
Done.
```

<u>total_payload_mass</u>
45596

Average Payload Mass by F9 v1.1

- Average payload mass = 2,928.4 kg
- Booster version F9 v1.1 typically carried moderate payload sizes.
- Represents early generation Falcon 9 performance capability.

```
%sql SELECT AVG("PAYLOAD_MASS__KG_") AS avg_payload_mass FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db  
Done.
```

<u>avg_payload_mass</u>

2928.4

First Successful Ground Landing Date

- Date: 2015-12-22
- Marks the first successful Falcon 9 ground landing.
- Major milestone in reusability development.
- Turning point in SpaceX operational strategy.

```
%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db  
Done.
```

MIN(Date)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
- These boosters successfully landed on drone ships.
- Payload mass range indicates mid-to-heavy missions.
- Demonstrates drone ship reliability for heavier payloads.

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_"
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Successful outcomes:
→ Total = 61
- Failure outcomes:
→ Total = 18
- Majority of landings were successful.

```
%sql SELECT TRIM("Landing_Outcome") AS Landing_Outcome, COUNT(*) AS count FROM SPACEXTABLE GROUP BY TRIM("Landing_Outcome") ORDER BY count DESC;
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	count
Success	38
No attempt	22
Success (drone ship)	14
Success (ground pad)	9
Failure (drone ship)	5
Controlled (ocean)	5
Failure	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Boosters Carried Maximum Payload

- Multiple F9 B5 boosters (e.g., B1048.4, B1049.4, B1051.3, etc.)
- Block 5 boosters carried the highest payload mass.
- Block 5 version optimized for heavy-lift and reusability.

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);
```

* sqlite:///my_data1.db
Done.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Month 01 – F9 v1.1 B1012 – CCAFS LC-40
- Month 04 – F9 v1.1 B1015 – CCAFS LC-40
- Both failures occurred in early 2015.
- Both used F9 v1.1 boosters.
- Occurred before consistent drone ship landing success.

```
%sql SELECT substr("Date",6,2) AS Month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE substr("Date",1,4) = '2015' AND "Landing_
```

```
* sqlite:///my_data1.db  
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Early years dominated by “No attempt”.
- Drone ship successes and failures were initially balanced.
- Shows transition period toward landing reliability.

```
%sql SELECT TRIM("Landing_Outcome") AS Landing_Outcome, COUNT(*) AS count FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY TRIM("Landing_Outcome")
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

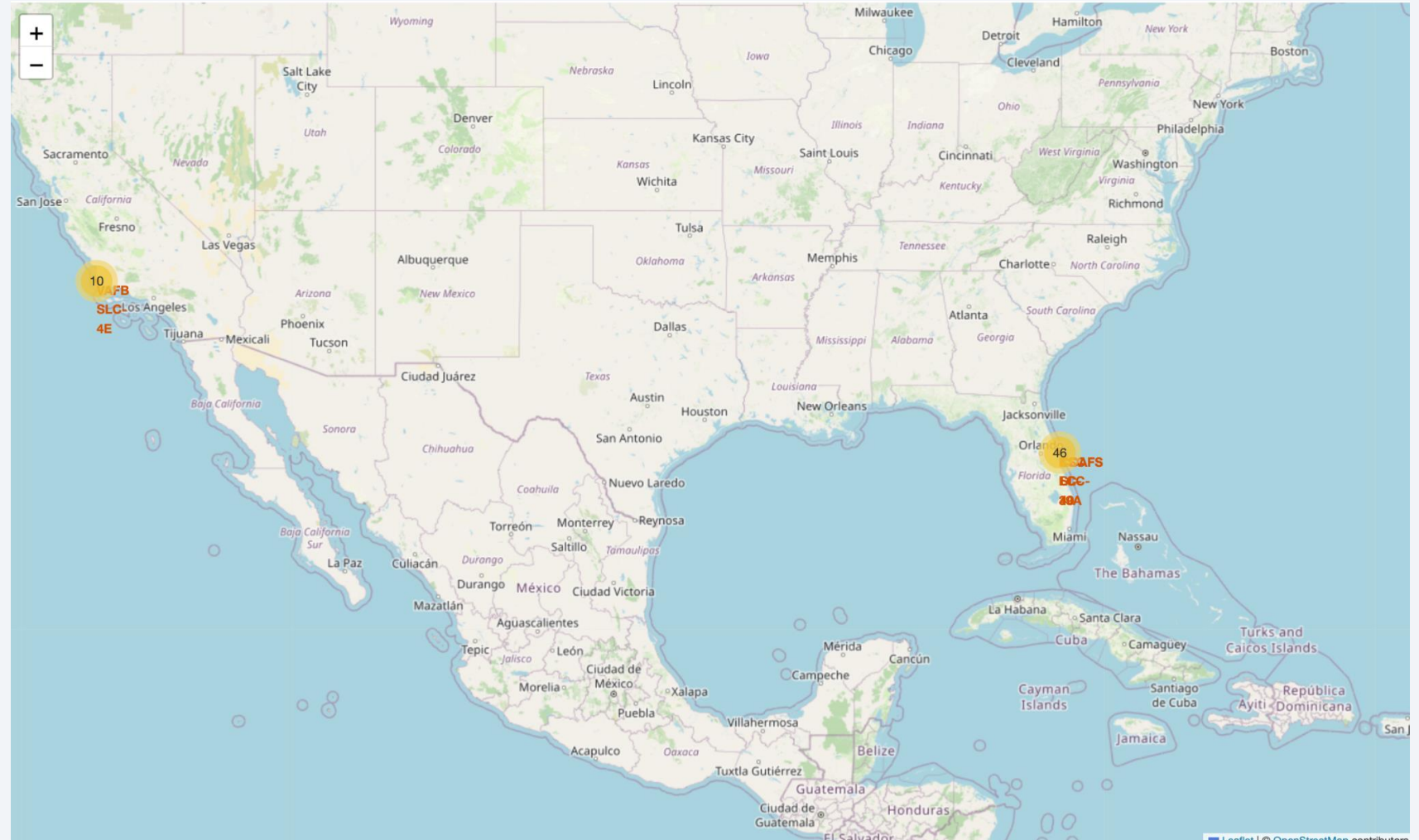
A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

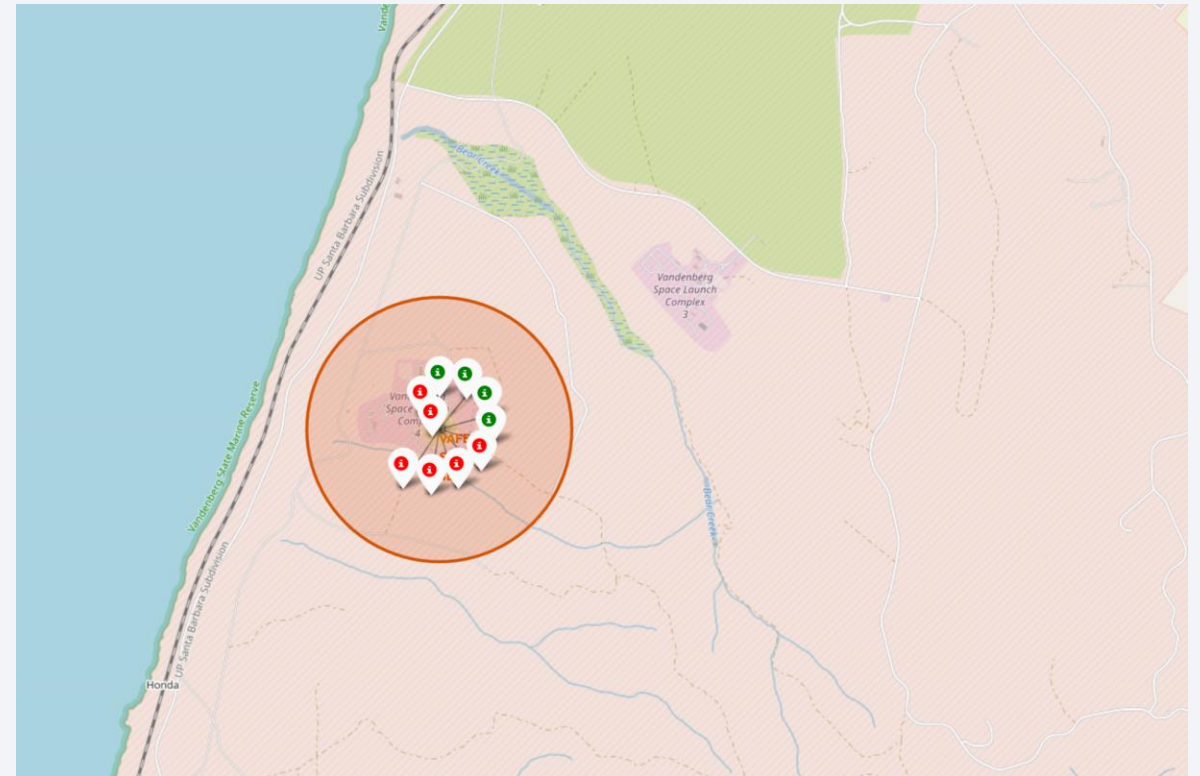
Geographical Locations of Falcon 9 Launch Sites

- Launch sites are located exclusively in the United States.
- Two major geographic clusters appear:
- Florida (CCAFS LC-40 and KSC LC-39A)
- California (VAFB SLC-4E)
- Florida hosts the majority of launches.
- Coastal locations allow safe launch trajectories over open ocean.



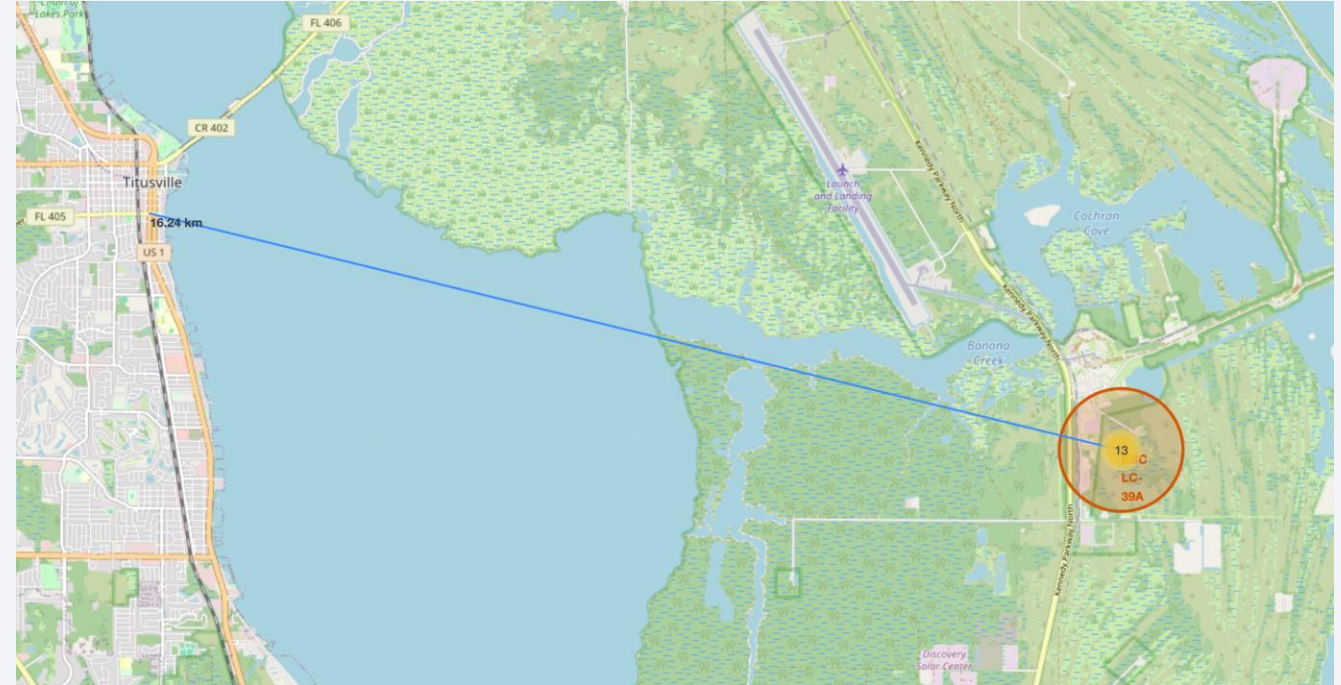
Color-Coded Launch Outcomes by Site

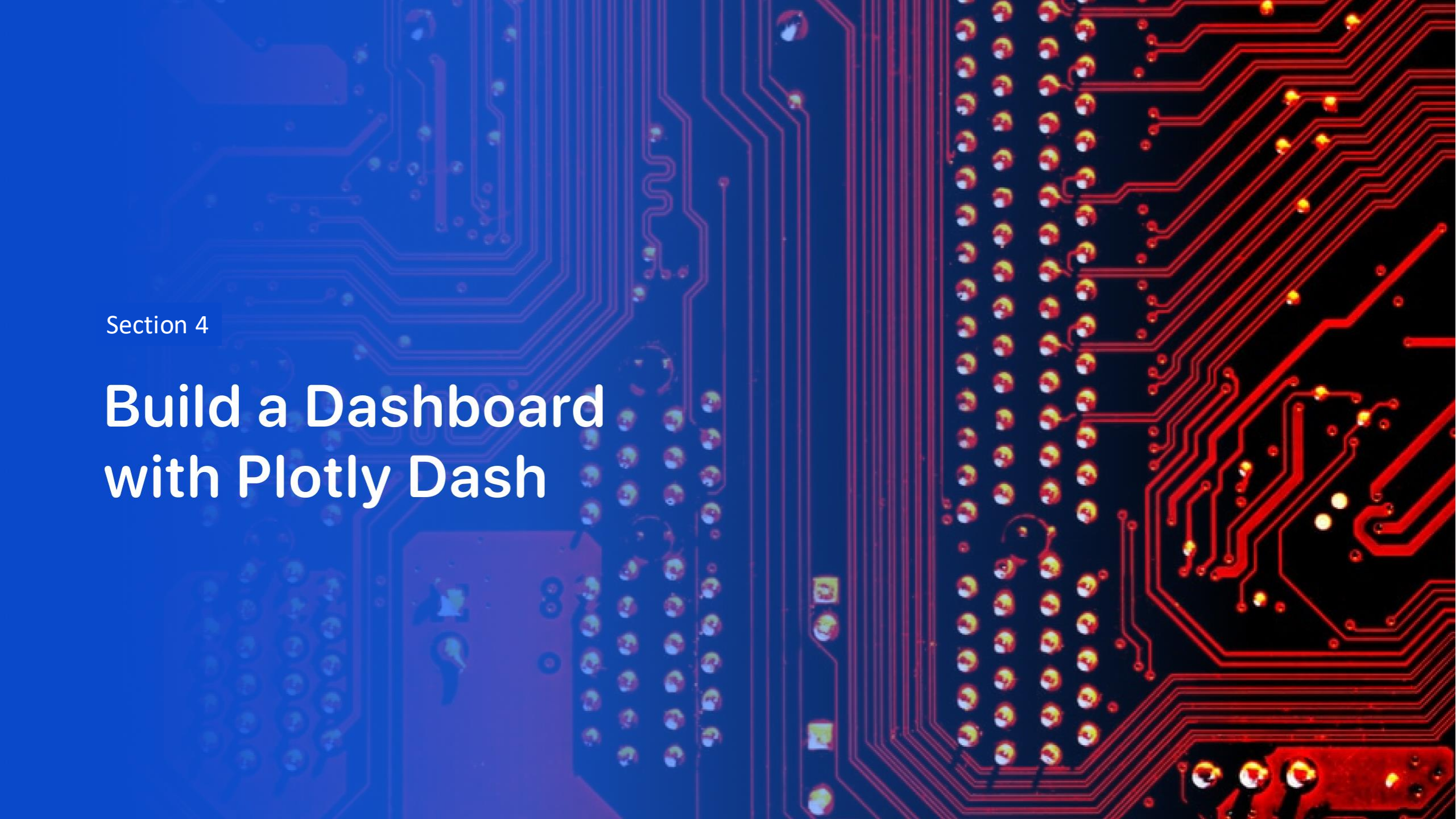
- Markers are color-coded to represent launch outcomes (green → successful landings; red → failed landings)
- Marker clustering groups multiple missions at the same launch site.
- The circular highlight indicates the operational radius around the launch complex.



Distance Analysis Between Launch Site and Nearby Infrastructure

- A distance line is drawn between the selected launch site (KSC LC-39A) and a nearby urban/infrastructure point (Titusville, FL), with the calculated distance (16.24 km) displayed directly on the map.
- The launch site is located at a safe distance from major urban areas.
- It is positioned close to the coastline, enabling safe eastward launch trajectories over the ocean.



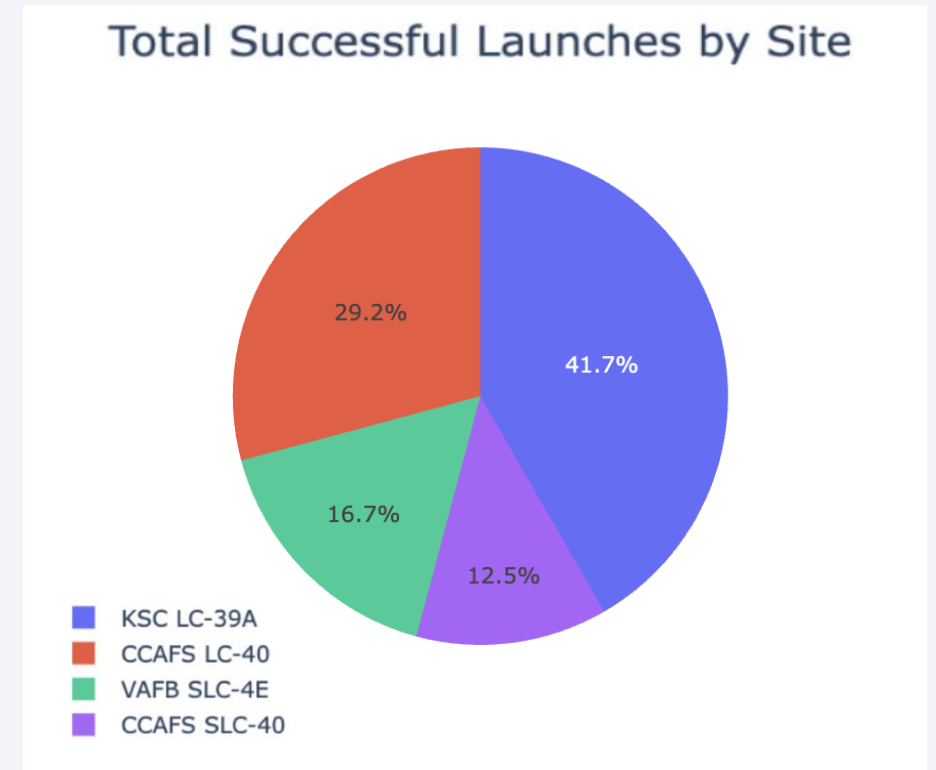


Section 4

Build a Dashboard with Plotly Dash

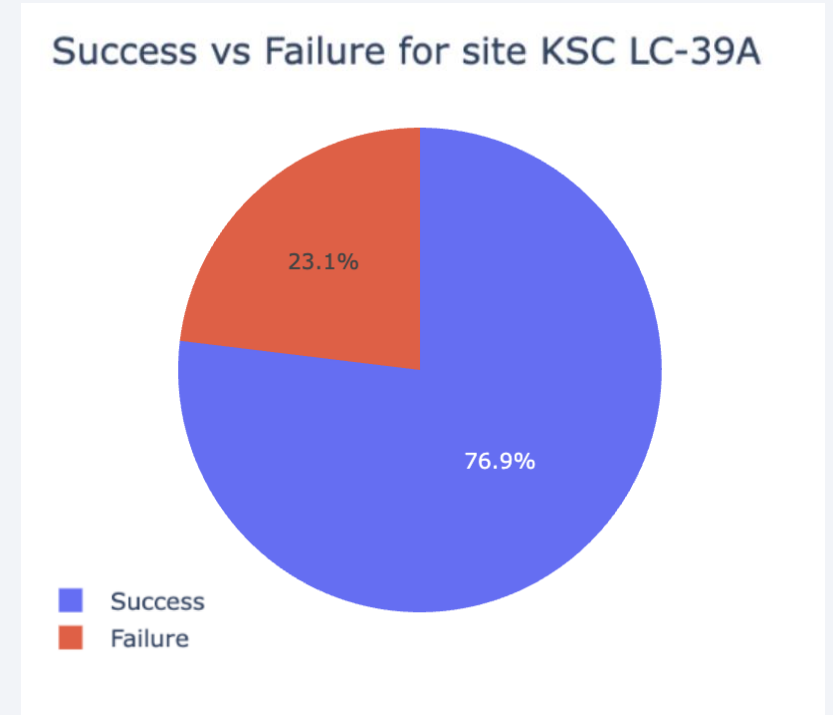
Distribution of Successful Launches Across Sites

- KSC LC-39A has the highest share of successful launches (~41.7%).
- CCAFS LC-40 follows with ~29.2%.
- VAFB SLC-4E and CCAFS SLC-40 contribute smaller shares.
- Launch success is concentrated primarily in Florida-based sites.
- This suggests Florida is the main operational hub for high-volume successful missions.



Success Ratio Analysis for KSC LC-39A

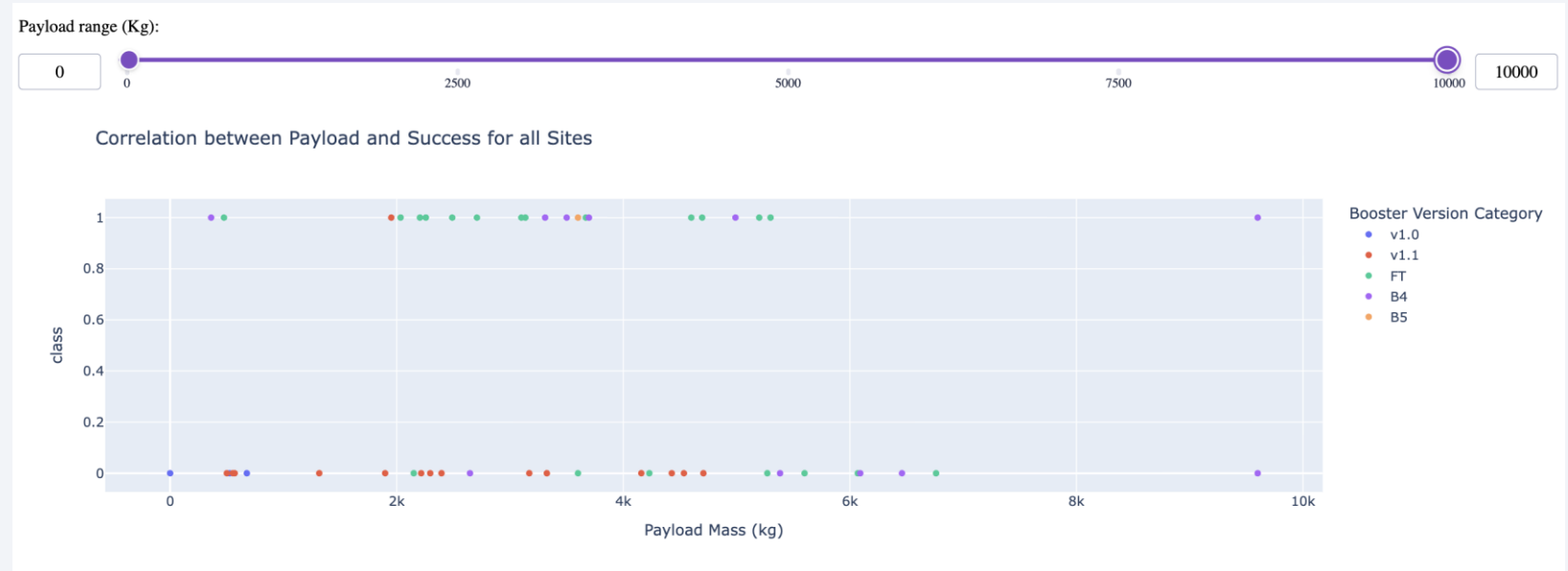
- KSC LC-39A has a 76.9% success rate, the highest among all launch sites.
- Only 23.1% of launches resulted in failure.
- The large success portion indicates strong operational reliability.
- This site appears to be the most efficient launch facility in the dataset.



Payload Mass vs Launch Outcome

- Payload vs Success Pattern

- There is no strict linear relationship between payload mass and success.
- Successful launches occur across a wide payload range (low to high).
- Failures are more frequent in lower-to-mid payload ranges.



- Booster Version Performance

- Newer booster versions (FT, B4, B5) show higher concentration of successful launches.
- Early versions (v1.0, v1.1) display more failures.

- High Payload Observations

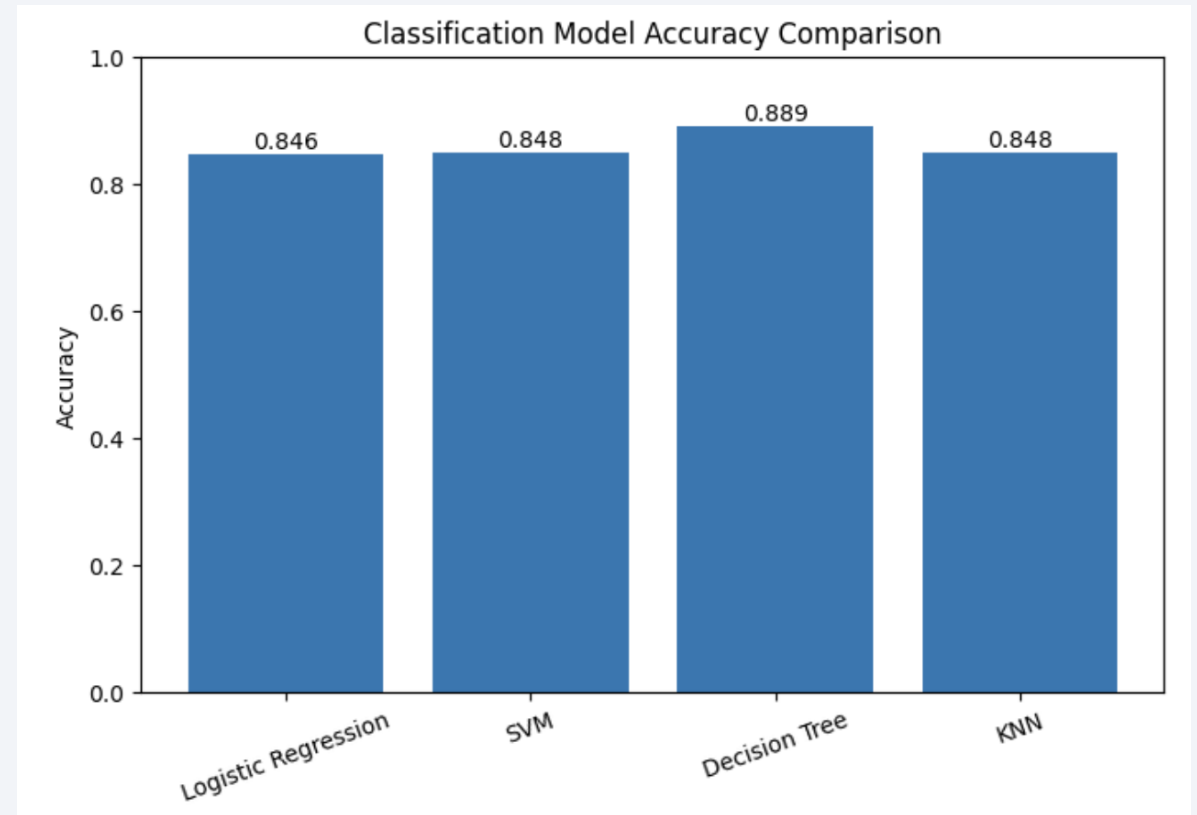
- Some of the highest payload missions (~9000–10000 kg) are successful.
- This suggests improved reliability over time.

Section 5

Predictive Analysis (Classification)

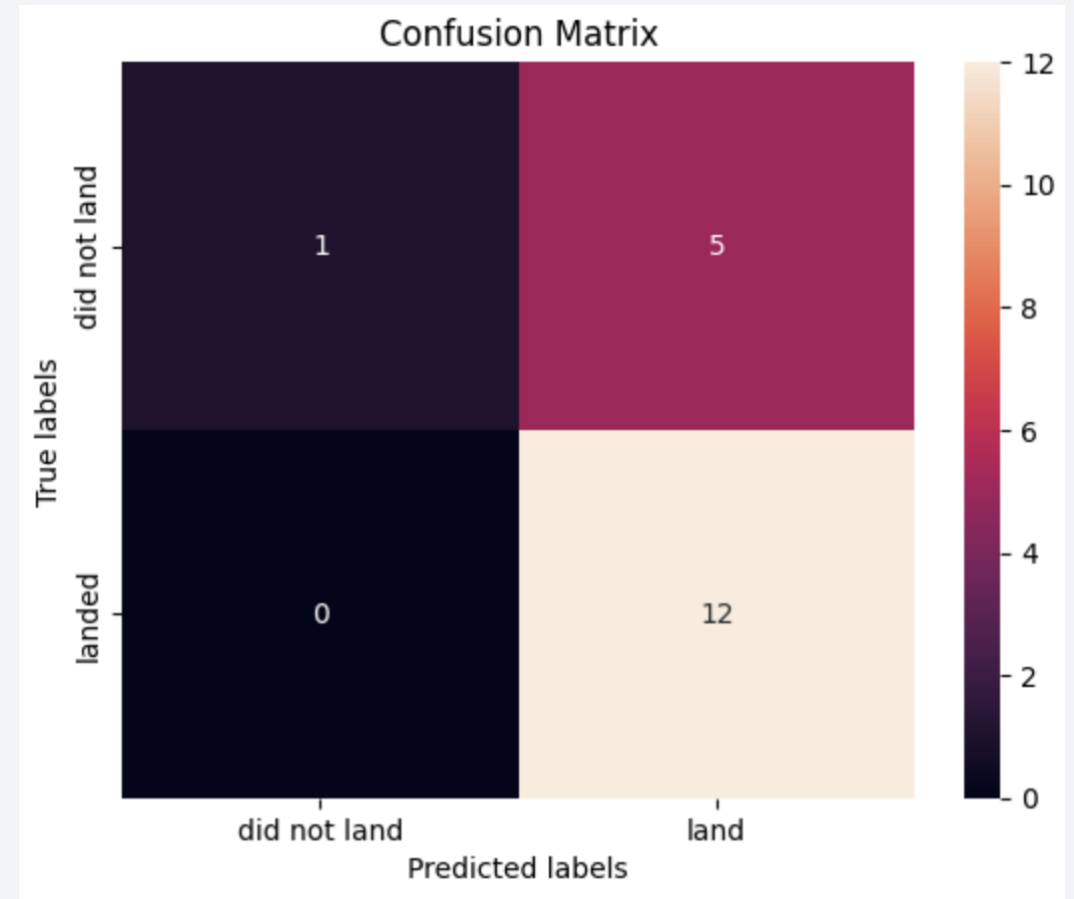
Classification Accuracy

- Decision Tree achieved the highest accuracy (0.889).
- Logistic Regression, SVM, and KNN performed similarly (≈ 0.846 – 0.848).
- Decision Tree outperforms the other models by about 4%.



Confusion Matrix

- 12 successful landings correctly predicted
- 5 failed launches misclassified as successful
- 0 missed successful landings
- Strong performance on “land” class
- Weak performance on detecting failures



Conclusions

- Launch success varies significantly by site, with certain sites showing higher success counts.
- Payload mass shows a relationship with launch outcome, particularly within specific booster versions.
- Among the tested models, Decision Tree achieved the highest cross-validation accuracy.
- However, confusion matrix analysis revealed class imbalance and weaker performance in identifying failed launches.
- Overall, the models are highly effective at predicting successful landings but less reliable at detecting failures.
- Further improvement could involve handling class imbalance or optimizing for recall on failed launches.

Appendix

Thank you!

